

Response to Reviewers — Round 2

Manuscript: Flux-Preserving Adaptive Finite State Projection
for Multiscale Stochastic Reaction Networks

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Dear Editor and Reviewers,

Thank you for the continued careful evaluation of our manuscript. We are grateful for the constructive feedback in this second round, and we have addressed each point raised by the referees. Below we respond point by point.

Response to Referee 1

Comment 1.1: *The “total flux” of a reduced state space is defined as the sum of the flux of each state in the subset. The majority of this total flux would then be the propensities to move between two states already contained in the reduced state space. If the goal of the algorithm is to avoid loss of states that move to other metastable regions, would it make more sense to limit the flux to propensities of moving outside the reduced state space?*

Response: The referee’s question is well taken. One could protect states based on their boundary outflux $\Phi_{\partial}(\mathbf{x}, t) := p(\mathbf{x}, t) \sum_{\mathbf{y} \in J'} \mathbf{A}_{\mathbf{y}\mathbf{x}}$ —the rate at which probability leaves $\mathbf{x}$ to states outside the current active set J —rather than total flux $\Phi(\mathbf{x}, t) = p(\mathbf{x}, t) w(\mathbf{x})$. However, total flux is preferable for two reasons that are most clearly illustrated by the bottleneck example.

Bottleneck states have near-zero boundary outflux. The defining feature of a bottleneck state s_b connecting metastable regions A and B is that its transitions lead to states *inside* those regions, which are already in J (they have high probability and survive the quantile filter). Consequently $\Phi_{\partial}(s_b, t) \approx 0$, and boundary-outflux protection would *not* retain s_b despite its structural importance. Total flux $p(s_b) w(s_b)$, by contrast, is large because $w(s_b)$ is large: it measures the rate at which probability *cycles through* s_b regardless of where those transitions lead. A large exit rate combined with small occupancy is precisely the hallmark of a fast-transit connector—and exactly the signature that boundary outflux misses once both endpoints are already in J .

Circular dependency. Computing $\Phi_{\partial}(\mathbf{x}, t)$ requires knowing J' , which is determined by the pruning decision itself. Total flux $p(\mathbf{x}) \cdot (-\mathbf{A}_{\mathbf{x}\mathbf{x}})$ is a byproduct of the generator diagonal and requires no knowledge of the pruning outcome.

We have added Remark 3.1 to the revised manuscript making this distinction explicit.

Comment 1.2 (minor): *Line 140: the term for flux $w(x)$ is used before being defined.*

Response: We have added an inline definition of $w(x)$ at its first use in Remark 2.3, so that the reader does not need to look ahead to Section 3 for the formal definition.

Comment 1.3 (minor): *Fig. 2: As the flux is central to your paper, it would be helpful to visualize it for the readers. A natural place would be in Fig. 2A, where the fluxes can be illustrated on the arrows.*

Response: We have updated Figure 2A to visualize probability flux on the arrows. Transitions incident on the bottleneck state s_b are now drawn with thicker lines to indicate that they carry high flux $\Phi(s_b, t) = p(s_b, t) w(s_b)$, while the internal metastable transitions (within regions A and B) are drawn thin to reflect their comparatively low flux. The label $\Phi(s_b) = p(s_b)w(s_b)$ appears on the representative $s_1 \leftrightarrow s_b$ arrow, and the annotation $p(s_b) \ll 1, w(s_b) \gg 0$ below s_b makes explicit the key property: s_b is rarely occupied yet transitions through it at a high rate, so its flux is non-negligible despite its low probability. This directly illustrates why probability-only pruning fails while flux-based pruning retains s_b .

Response to Referee 2

Theoretical Issues

Comment 2.1: *There is a significant notational error in Eq. (4.4) regarding the definition of boundary fluxes. Given the stated column convention, Φ_{out} should involve $\mathbf{A}_{J'J}$ rather than $\mathbf{A}_{JJ'}$.*

Response: We thank the referee for catching this. We have added an explicit clarification of the column convention immediately after Eq. (4.4): the block $\mathbf{A}_{J'J}$ has rows indexed by J' and columns indexed by J , so that the product $\mathbf{A}_{J'J} \mathbf{p}_J$ gives the rate at which probability flows *out* of J into J' . This convention is now stated in the text and the notation is consistent throughout Section 4.

Comment 2.2: *The derivation for the adaptive time-step rule ($\Delta t = \varepsilon_{\Delta t} / \Phi_{\text{max}}$) relies on the inequality $\Phi_{\text{in}} + \Phi_{\text{out}} \leq \Phi_{\text{max}}$, for which I fail to see the justification without a multiplicative constant that accounts for the boundary's size or structure.*

Response: The referee is correct that the original inequality $\Phi_{\text{in}} + \Phi_{\text{out}} \leq \Phi_{\text{max}}$ was not justified. We have replaced it with the trivially valid bound $\Phi_{\text{out}} \leq \Phi_{\text{total}}$, which holds because the boundary outflux is a subset of the total system flux. The adaptive time-step rule in Corollary 4.7 now reads $\Delta t \leq \varepsilon_{\Delta t} / \Phi_{\text{total}}$, and the proof follows immediately from this containment. This correction propagates cleanly through the subsequent corollaries without affecting the structure of the error bounds.

Comment 2.3: *The algorithm (3.1 and 3.3) performs renormalization after pruning, yet the theoretical analysis in Section 4 is built upon a compressed (reflecting) generator. The relationship between these two treatments needs explicit clarification to ensure the error bounds remain valid for the implemented code.*

Response: We appreciate this question, as it highlights a point that deserved clarification. Upon reflection, there is in fact *no gap* between the theory and the implementation, because the implementation directly constructs the compressed generator—not the truncated generator followed by renormalization.

Specifically, Algorithm 3.2 (forward-enumeration matrix construction) builds the generator column by column. For each source state $x \in J$ and each reaction k , the propensity $\alpha_k(x)$ contributes to the off-diagonal entry at row $y = x + \nu_k$ *only if* $y \in J$. The diagonal entry for column x is then set to the negative of the sum of these included off-diagonal entries. Transitions whose destination falls outside J are excluded from both the off-diagonal and the diagonal. Consequently, the column

sums are exactly zero by construction. This is precisely the defining property of the compressed generator $\tilde{\mathbf{A}}_{JJ}$ (Proposition 4.1, property 1), and the matrix exponential $e^{\tilde{\mathbf{A}}_{JJ}\Delta t}$ therefore conserves $\mathbf{1}^\top \mathbf{p}$ exactly.

The renormalization $\mathbf{p} \leftarrow \mathbf{p}/\|\mathbf{p}\|_1$ in Algorithm 3.1 (Stage 3) serves a *different* purpose: it compensates for the probability mass discarded when low-probability states are removed during pruning, not for any leakage from the generator. The pruned mass is at most α by the quantile selection in Stage 1.

We have added Remark 4.4 to the revised manuscript to make this distinction explicit, stating that the implementation uses the compressed generator and that the renormalization addresses pruning losses only. The error analysis in Section 4, which is formulated entirely in terms of $\tilde{\mathbf{A}}_{JJ}$, therefore applies directly to the implemented algorithm without any additional approximation gap.

Comment 2.4: *Estimating $w_{\max}$: Corollary 4.7 uses $w_{\max}$ (the maximum exit rate of pruned states), but it is unclear how this value is robustly estimated on a truncated domain since it involves states outside the current active set.*

Response: We have clarified this in the revised manuscript. The key observation is that the exit rate $w(x) = \sum_k \alpha_k(x)$ is computed for every state x in the active set *before* pruning takes place (it is needed for the flux computation in Algorithm 3.1, Stage 2). Since pruning only removes a subset of the current active states, $w_{\max}$ over the pruned states is available as a byproduct of the flux computation, with no need to evaluate $w(x)$ at states outside the active set. Corollary 4.7 now states this explicitly.

Empirical Issues

Comment 2.5: *The manuscript lacks head-to-head comparisons against strong adaptive FSP baselines, such as ETFSP or other time-varying variants.*

Response: We have added Section 5.6 to the revised manuscript, which provides a head-to-head comparison of the flux-based adaptive FSP against the Krylov-FSP-SSA method of Vo and Sidje [Sidje2015], reviewed by Dinh and Sidje [Dinh2016]. This baseline shares the same expand-evolve-prune skeleton and Krylov matrix exponential integrator, differing in pruning (joint probability-and-derivative threshold per Algorithm 2 of [VoSidje2016], without flux-based protection), time stepping (mass-conservation accept/reject rather than flux-adaptive), and expansion (SSA-driven with reachability enumeration). The comparison isolates the effect of flux-aware pruning and flux-adaptive time stepping on three benchmark systems.

On the Oregonator, the flux-adaptive time stepping provides a decisive advantage: the adaptive FSP completes the simulation in 36s using 26 snapshots with $|S| \leq 1,057$, while the Krylov-FSP-SSA method reaches only $t = 0.003$ (2.7% of the time horizon) after 3,000 iterations and 74s, with the state space growing to 13,294 states. On the bottleneck system, both methods correctly retain the connector state $B=1$ (the joint probability-and-derivative filter keeps it since $p(B=1) \gg \text{droptol} = 10^{-10}$), but Krylov-FSP-SSA produces $\langle C \rangle = 0.12$ ($40\times$ below the reference $\langle C \rangle = 4.978$) because its step size grows exponentially ($\tau : 100 \rightarrow 3,700\text{s}$) and the depth- $r=1$ expansion cannot track the probability wavefront spreading over $C \in [0, k_2 t_f]$; FP-FSP avoids this by using a fixed step $\delta t = 1/k_2 = 10\text{s}$. On the Robertson system over $[0, 10^4]$, only FP-FSP completes the horizon: the stiffness of $k_2 = 3 \times 10^6$ locks Krylov-FSP-SSA's step size in $[3 \times 10^{-4}, 3 \times 10^{-3}]$, reaching only $t = 228$ (2.3%) in 500 iterations.

These results are presented in Tables 7–9 and Figures 7–8 of the revised manuscript

Comment 2.6: *The authors should attempt to numerically demonstrate the specific benefits of flux-based protection versus pure probability pruning, and of $\Phi_{\max}$ -based stepping versus other activity*

indicators.

Response: The comparison in the new Section 5.6 directly demonstrates these benefits. The Oregonator provides two complementary illustrations. First, flux-based pruning keeps the active set compact ($|S| \leq 1,057$) because the flux criterion identifies and discards dynamically irrelevant states; the joint probability-and-derivative filter used by Krylov-FSP-SSA conservatively retains all states with significant positive derivative (expanding boundary states), causing the projection to grow to 13,294 states without bound. Second, Φ_{total} -based stepping adjusts Δt from 1.6×10^{-6} to 1.9×10^{-5} to match instantaneous system activity, while the accept/reject mechanism is locked at $\tau \approx 10^{-6}$ because the large propensities cause mass loss for any larger step. The bottleneck system demonstrates the step-size control advantage: both methods retain $B=1$ (the joint filter keeps it since $p(B=1) \gg \text{droptol}$), but FP-FSP’s fixed step $\delta t = 1/k_2$ keeps expansion pace-matched with the physical probability wavefront, yielding $\langle C \rangle = 4.978$ ($< 0.1\%$ error), while Krylov-FSP-SSA’s exponentially growing τ produces a 14-state projection that misses the wavefront, yielding $\langle C \rangle = 0.12$ ($40\times$ error). Quantitative metrics are reported in Tables 7–9

Comment 2.7: *Some guidance for choosing the key tolerances ($\alpha, \varepsilon_{\text{flux}}, \varepsilon_{\Delta t}$) across different types of models would be very valuable.*

Response: We have added a parameter guidance table (Table 9 in the revised manuscript) and Remark 4.9 providing concrete parameter guidance. The table covers four qualitatively different system types (connectivity-critical, oscillatory, stiff autocatalytic, and bimodal/metastable), with the parameters used in the corresponding numerical experiments shown in parentheses for direct reference. Three practical rules emerge from the benchmark experience and the theoretical framework of Remark 4.8.

- $\varepsilon_{\Delta t}$: Choose $\varepsilon_{\Delta t} \leq 1/\kappa$ so that depth-one expansion suffices (Corollary 4.7); for $\kappa \approx 1$ this permits $\varepsilon_{\Delta t}$ up to 1. Reduce $\varepsilon_{\Delta t}$ to 0.01–0.1 when a long integration horizon demands high accuracy, since local errors accumulate additively (Corollary 4.4).
- α : For connectivity-critical systems, α can be set aggressively (0.7–0.9) because the flux criterion rescues any misclassified connector states; the quantile filter then removes only the genuinely irrelevant tail. Without bottlenecks, moderate values (0.1–0.3) avoid over-pruning.
- $\varepsilon_{\text{flux}}$: Set to zero for oscillatory or well-mixed systems where the active set is simply connected (a positive value would retain unnecessary states near the expanding boundary). For bottleneck systems, match $\varepsilon_{\text{flux}}$ to the expected relative flux of the connector state: $\varepsilon_{\text{flux}} \sim p(s_b)w(s_b)/\Phi_{\text{total}}$; in practice 10^{-12} – 10^{-6} covers most cases.

Summary

We thank both referees for their continued careful reading. All outstanding concerns have been addressed.

On the theoretical side, Referee 2’s notational error in Eq. (4.4) and the unjustified time-step inequality have been corrected, the relationship between the implemented renormalization and the compressed generator has been made explicit (Remark ??, Section 3.3), and the availability of w_{max} at pruning time has been clarified (Corollary ??).

On the empirical side, the new Section ?? provides a head-to-head comparison of the flux-adaptive FSP against the Krylov-FSP-SSA method on three benchmark systems, directly demonstrating the benefits of flux-based pruning (bottleneck system) and flux-adaptive time stepping (Oregonator).

In response to Referee 1’s conceptual question, Remark 3.1 explains why total flux is the correct protection criterion: boundary outflux is approximately zero for genuine bottleneck states (because their transitions lead into the already-retained high-probability regions) and is in any case circular to compute during the pruning step.

Finally, the new parameter guidance table (Table ??) and Remark ?? address Referee 2’s request for parameter guidance, giving concrete recommended ranges for $(\alpha, \varepsilon_{\text{flux}}, \varepsilon_{\Delta t})$ across four qualitatively different model types, grounded in both theory and the benchmark experiments.

Sincerely,

Aditya Dendukuri, Shivkumar Chandrasekaran, and Linda Petzold